

Cognitive Socially Assistive Robot for Multimodal Assessment and Adaptive Support in Autism Therapy

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Abstract: Intervention for social communication deficits, joint attention difficulties, and emotional dysregulation in Autism Spectrum Disorder (ASD) requires constant, individualized, and data-informed intervention. This is typically an area where traditional clinical models may fail. In this paper, we aim to describe a Cognitive Socially Assistive Mobile Robot (C-SAR), that focuses on improving therapeutic outcomes, through multimodal assessment and interactive engagement. The system- C-SAR, is built on a TurtleBot3 platform, and includes eye gaze tracking analysis, facial expression recognition analysis, and speech analysis, allowing it to calculate three key metrics of behavior, Joint Attention Score (JAS), Social Engagement Score (SES) and Speech Clarity Score (SCS). During intervention with the robot, children with ASD engage in adaptive, game-like tasks using real-time audiovisual feedback, with the robot providing positive reinforcement to strengthen attention, imitation, and social reciprocity. Leveraging on-device deep learning inference, the robot autonomously adapts prompts based on real-time behavioral cues while a therapist-facing dashboard provides longitudinal visualization. Preliminary validation demonstrates 78-85% speech imitation accuracy, 85-90% gaze fixation success rates, and notably increased willingness to vocalize compared to passive listening tasks. This work contributes a mobile, emotionally engaging, and clinically interpretable robotic platform that augments traditional therapy for children with ASD.

Keywords: Autism Therapy, Assistive Robotics, Gaze Tracking, Emotion Recognition, ROS 2, Human-Robot Interaction

I. INTRODUCTION

Autism Spectrum Disorder (ASD) is a complex neurodevelopmental condition characterized by persistent challenges in social communication, joint attention, restricted interests, and repetitive behaviors. Affecting approximately one in every 100 children worldwide, ASD presents wide heterogeneity across cognitive, behavioral, and emotional domains [1]–[3]. Children with ASD often struggle with eye contact, recognizing emotions, imitating gestures, and engaging in reciprocal communication—skills essential for social learning and emotional regulation [4]. Numerous studies show that early, intensive behavioral interventions

improve language, emotional understanding, and adaptive functioning [5]. However, traditional one-on-one therapy models are resource-intensive, time-consuming, and difficult to scale, especially in underserved clinical settings. These approaches also rely on subjective observation, leading to potential variability and bias in assessing therapeutic progress [6], [7]. To address these challenges, Socially Assistive Robots (SARs) have emerged as promising tools that combine automation, consistency, and engagement in therapy [8]–[10]. Unlike robots designed for physical assistance, SARs focus on cognitive and social interaction, offering predictable, emotionally neutral communication styles suited to children with ASD [11]. Studies show that autistic children often display greater attention and motivation when interacting with robots, as robotic predictability reduces the social-cognitive load associated with human interaction [12], [13]. By delivering consistent, non-judgmental feedback, SARs effectively support imitation, joint attention, and emotional expression [14]. Moreover, their ability to provide multimodal feedback—visual, auditory, and tactile—makes them valuable for reinforcing learning and objectively tracking engagement [15], [16]. Despite these benefits, current SAR implementations face several limitations. Humanoid robots such as NAO, KASPAR, and Pepper show therapeutic promise but remain costly, fragile, and difficult to maintain [17]–[19]. Non-humanoid mobile platforms like TurtleBot offer a low-cost, portable alternative that is less overwhelming for children [20], [21]. However, most SARs still rely on unimodal perception, analyzing gaze, emotion, or speech independently rather than as integrated behavioral data. This approach overlooks the dynamic interdependence of cognitive and affective states that define social behavior [22], [23]. Furthermore, static, pre-scripted interactions limit personalization, while the absence of clinical analytics restricts interpretability for therapists [24]. Although recent advances in Artificial Intelligence (AI) have enabled multimodal behavioral analysis through computer vision, speech, and affective computing [25], few systems incorporate adaptive reasoning—the ability to modify behavior in real time based on user engagement [26]–[28].

This disconnect between AI-driven perception and cognitive adaptation highlights the gap between technological sophistication and therapeutic utility.

In summary, existing SARs remain fragmented, non-adaptive, and underutilized due to high costs, unimodal sensing, and the absence of standardized engagement metrics [29], [30]. To overcome these barriers, this paper proposes a Cognitive Socially Assistive Robot (C-SAR)—an affordable, AI-powered, multimodal framework capable of real-time behavioral assessment through gaze tracking, facial emotion recognition, and speech analysis. By computing and integrating the Joint Attention Score (JAS), Social Engagement Score (SES), and Speech Clarity Score (SCS), the C-SAR system establishes an adaptive feedback loop that bridges robotic perception with human-centered, data-driven therapy.

II. METHODOLOGY

Addressing social communication, joint attention, and emotional challenges in children with Autism Spectrum Disorder (ASD) requires a replicable, individualized, and data-driven therapeutic framework—qualities often difficult to sustain in traditional human-led interventions. The proposed Cognitive Socially Assistive Robot (C-SAR) employs an AI-based perception–cognition–emotion (PCE) model that delivers adaptive, emotionally responsive therapy. By interpreting gaze, facial emotion, and speech in real time, AI acts as a mediator of engagement, transforming therapy from scripted routines into context-aware, interactive sessions that sustain motivation and participation.

The C-SAR framework integrates adaptive cognition and multimodal perception into a closed-loop model where the robot interprets behavior dynamically and adjusts its strategy accordingly. Through gaze tracking, emotion recognition, and speech analysis, it computes three engagement metrics—Joint Attention Score (JAS), Social Engagement Score (SES), and Speech Clarity Score (SCS)—to quantify attention, emotion, and verbal clarity. Each behavior triggers immediate feedback, reinforcing learning through cause-and-effect interaction that makes therapy rewarding and game-like.

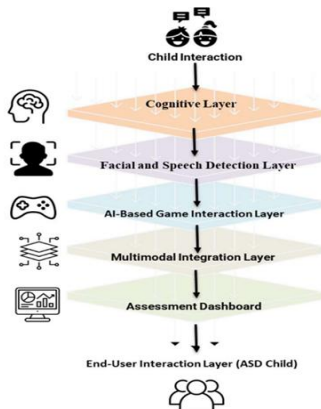


Figure1: Six-layer Cognitive–Therapeutic Framework integrating sensing, cognition, and adaptive feedback for personalized autism therapy.

As shown in Fig. 1, the C-SAR framework employs a six-layer cognitive–therapeutic architecture that seamlessly links perception, cognition, and feedback. The Cognitive Layer

manages adaptive decision-making through reinforcement learning, while the Facial and Speech Detection Layer interprets emotional and communicative cues. The AI-Based Game Layer delivers gaze-controlled and speech-imitation tasks that adjust pacing and difficulty to sustain engagement. The Multimodal Integration Layer fuses vision, audio, and physiological data to compute engagement metrics—JAS, SES, and SCS—for real-time analytics. The Therapy Dashboard Layer visualizes progress and engagement trends, and the End-User Layer provides expressive, emotionally safe interactions that build trust and confidence.

Together, these layers form a closed-loop adaptive framework that continuously transforms behavioral input into personalized feedback. By combining AI reasoning, multimodal sensing, and therapist supervision, C-SAR functions as an intelligent therapeutic companion on a TurtleBot3 platform equipped with an Intel RealSense camera, microphone array, and Raspberry Pi 4 for real-time perception, cognition, and adaptive feedback.

III. SYSTEM ARCHITECTURE

The Cognitive Socially Assistive Robot (C-SAR) described here is there is a highly mobile, flexible, mobile intelligent platform borrowing elements from multimodal perception, adaptive cognition, and interactive therapeutic modules under the same intelligent cognitive umbrella. This system operates on the following architecture: a bottom up sensory system which integrates both environmental and behavioral signals from the child, which affect, in real time, how the robot will behave; and a higher-order top-down cognitive system for adaptively and functionally engaging a child with Autism Spectrum Disorder (ASD) in alternative therapeutic experiences.

A. Hardware Framework: Mobile and Multimodal Sensing

The proposed hardware framework uses the TurtleBot3 Burger for its compact, modular design and compatibility with ROS 2. Its mobility allows dynamic positioning during joint-attention tasks for children with Autism Spectrum Disorder (ASD). An Intel RealSense D435i camera enables real-time RGB–D facial and gaze tracking, while a USB microphone array captures speech and localizes sound. An onboard speaker provides prompts, cues, and verbal rewards, and a Raspberry Pi 4 (8 GB) handles AI inference and sensor fusion, ensuring fully autonomous, real-time operation without external computing.



Fig. 2. C-SAR hardware framework integrating the camera and microphone as well as the onboard processor.

1) Joint Attention Score (JAS) – Frogle Level-1 (Engagement Module)

The Frogle Level-1 Engagement Game trains early eye-gaze awareness and visual attention in children with Autism Spectrum Disorder (ASD). A cartoon frog rhythmically hops between lily pads with sound cues (Fig. 3), while the system's camera continuously tracks the child's gaze to measure attentional focus on the display.

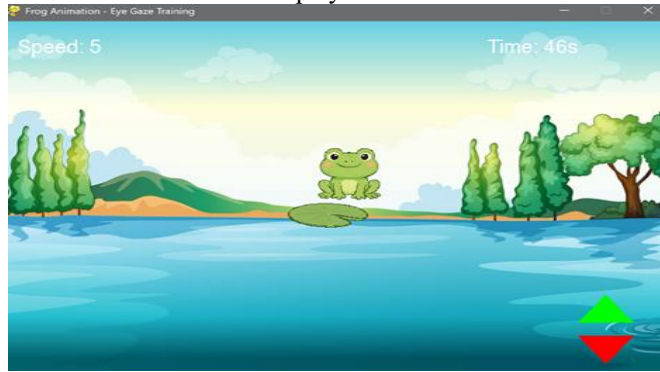


Figure 3: Frog animation interface for eye-gaze training (Frogle Level-1).

Working Principle

The module uses Pygame for interactive visuals and OpenCV Haar-cascade classifiers to detect facial and eye regions. Each frame is processed in grayscale, generating a binary signal that indicates real-time eye presence:

$$G(t) = \begin{cases} 1, & \text{if } \max(H_e(x,y)) > 0 \\ 0, & \text{otherwise} \end{cases} \dots\dots\dots (1)$$

where $H_e(x,y)$ is the Haar-cascade response for eyes. $G(t)=1$ indicates a "Tracked" state (eyes visible), and $G(t)=0$ indicates gaze loss..

Game Timing and Design

At initialization, frog and lily-pad sprites are randomly positioned on the background. The frog's motion follows a parabolic path

$$y(t) = y_s + (y_e - y_s)t - A(1 - (2t - 1)^2) \dots\dots\dots (2)$$

Where (x_s, y_s) and (x_e, y_e) are represent the jump coordinates, $t \in [0,1]$ is the normalized time, and $A=50$ pixels defines the jump amplitude.

Therapeutic Significance

Children Frogle Level-1 uses structured motion and gaze detection to help children with ASD sustain attention.

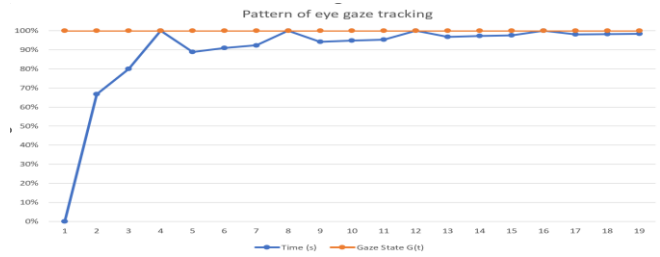
$$P_{tracked} = \frac{\sum_t G(t)}{N} \times 100\% \dots\dots\dots (3)$$

where (N) is the total frame count. $P_{tracked}$ represents the proportion of frames.

Relation of Equations to Results

Equations (1)–(3) form the basis for engagement analysis, with pilot results showing JAS values of 70–85%, confirming that rhythmic motion and accurate gaze detection promote sustained visual attention.

Figure. 4 Pattern of eye-gaze tracking for a 60 s Frogle Level-1 session.



2) Social Engagement Score (SES) – Frogle Level-2 (Interactive Module)

The Social Engagement Score (SES) module extends the passive gaze-tracking task of Level 1 by adding an eye-gaze-controlled interaction that introduces an element of decision-making and intentional engagement. In this level, the frog remains stationary until the child deliberately focuses on the next lily pad target for a fixed dwell time. Once sustained gaze is detected, the frog jumps, producing a direct cause-and-effect link between the child's attention and the visual response. This interaction reinforces sustained attention, anticipatory control, and visual-motor coordination in children with autism spectrum disorder (ASD).

As shown in Fig. 5, the gaze-controlled environment with animated frog but transforms the visual experience into a responsive, attention-dependent system.

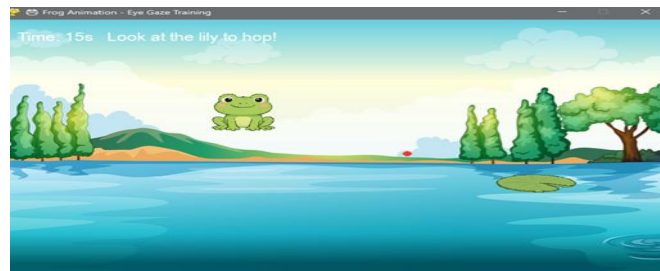


Figure 5: Eye-gaze-controlled interaction interface for Frogle Level-2. The frog leaps only when the child maintains gaze fixation on the target for the required dwell time.

1) Working Principle

The The system uses an OpenCV-based gaze-estimation engine that detects the child's face and eyes, computes pupil centroids (cx,cy) , and normalizes them to generate real-time gaze coordinates:

$$x_g = \frac{x + e_x + c_x}{W_f} \dots\dots\dots (4)$$

$$y_g = \frac{y + e_y + c_y}{H_f} \dots\dots\dots (5)$$

where (x, y) denotes the top-left corner of the detected face, e_x, e_y the offset of the eye region, and W_f, H_f the frame dimensions.

These normalized gaze coordinates are mapped to the display resolution:

$$X_s = x_g \cdot W_s \dots\dots\dots (6)$$

$$Y_s = y_g \cdot H_s \dots\dots\dots (7)$$

where W_s, H_s denote screen width and height.

When the child's gaze (x_s, y_s) If gaze stays on the target longer than T_d , the frog jumps and the fixation event is recorded:

$$D(t) = \begin{cases} 1, & \text{if } \int_{t_0}^t [P_g(\tau) \in R_L] d\tau \geq T_d \\ 0, & \text{otherwise} \end{cases} \quad (7)$$

where $P_g(\tau)$ represents the gaze position at time τ and R_L the lily pad region of interest. $(D(t)=1)$ indicates a valid fixation, triggering the frog's jump. The reaction time for each fixation event is computed as:

$$RT = t_{\text{jump}} - t_{\text{spawn}} \quad (8)$$

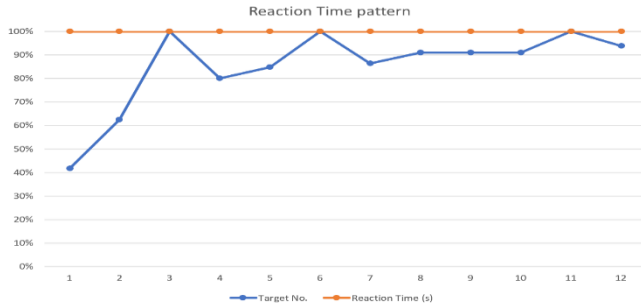


Figure 6. Reaction time pattern across successive gaze-triggered jumps in Frog Level-2.

3) Speech Clarity Score (SCS) – Monkey Game (Speech-Based Imitation and Verbal Interaction Module)

The Monkey Game module focuses on verbal imitation, auditory attention, and speech initiation—skills often underdeveloped in children with ASD. Unlike the non-verbal Frog tasks, it introduces interactive vocal play with a cartoon monkey that prompts imitation (e.g., “Say sheep!”), rewarding correct responses with animations and encouraging voice feedback, while missed attempts prompt gentle repetition.

This is a Cat



Figure 7: The Monkey Game interface for speech-based imitation and verbal engagement training in children with ASD.

1) Working Principle

The game integrates an AI-driven speech recognition and feedback loop using the Speech Recognition, Ollama LLM, and pytsx3 libraries.

Speech Recognition converts live audio from the USB microphone array into text via the Google ASR engine.

Each trial follows a structured listen → speak → evaluate cycle. When the child responds, the module computes Speech Accuracy (A) as:

$$A = \frac{\text{Match}(S_r, S_t)}{\text{len}(S_t)} \times 100\% \quad (11)$$

where S_r is the recognized speech, S_t the target phrase, and ‘Match ()’ counts correctly aligned characters or phonemes. If $(A > 70\%)$, the system classifies the response as correct and plays positive reinforcement feedback (“Great job!”). Otherwise, the monkey encourages retry with simplified phrasing or exaggerated pronunciation.

2) Game Design and Parameters

Each session lasts 60 seconds, divided into 5–10 imitation trials. Prompt phase: The monkey delivers a spoken instruction (~2 s).

Response window: The system listens for up to 5 s after prompt completion.

Feedback phase: Immediate audiovisual reinforcement based on recognition results.

Parameter	Symbol	Value
Session Duration		60 s
Average Prompt Cycle		5 s
Voice Output Rate		180 wpm
Recognition Threshold	A	70 %
Feedback Delay		≤ 0.3 s
Trials per Session	N	5–10

The difficulty level can be dynamically adjusted:

- Beginner mode: Single-word prompts (“Cat”, “Dog”).
- Intermediate: Two-word phrases (“Say Apple”, “Touch Nose”).
- Advanced: Simple sentences (“Can you say Hello Monkey?”).

This flexibility allows therapists to tailor speech complexity and pacing to individual cognitive and linguistic profiles.

Children exhibited improved willingness to vocalize compared to passive listening activities. The positive reinforcement cues (animated monkey cheers and praise voice) significantly boosted attention span and participation. Sessions with slower pacing benefitted children with delayed articulation, while faster cycles enhanced spontaneous responding and conversational rhythm..

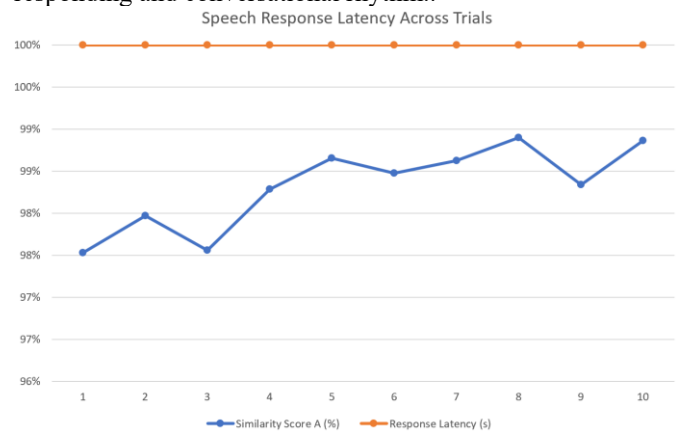


Figure 8: Speech imitation performance across successive trials in the Monkey Game. Higher accuracy and reduced latency demonstrate improved verbal initiation and adaptive learning.

The quantitative metrics Speech Accuracy (Eq. 12) and Response Latency (Eq. 13) are sound measures of verb output. The speech accuracy (78–85%) and the latency measure being lower than prior modules indicates that the C-SAR used in the module promotes articulation as well as confidence and motivation to communicate verbally.

This level builds upon the visual attention activities of Froggle Level-1 and Level-2, and by having a multimodal level experience, the Cognitive Socially Assistive Robot (C-SAR) can assess and support communication across both verbal and non-verbal domains during ASD therapy.

IV. OVERALL RESULTS AND ANALYSIS

A comprehensive evaluation of the proposed Cognitive Socially Assistive Robot (C-SAR) was performed across three interactive therapeutic modules Froggle Level-1 (JAS), Froggle Level-2 (SES), and Monkey Game (SCS)—each addressing a different cognitive-behavioural domain. Collectively, these three modules enabled us to measure visual attention, social reciprocity, and speech imitation to develop a complete behavioural profile of engagement in children with autism spectrum disorder (ASD).

The comparative results are presented in Table I.

Module	Target Cognitive Skill	Metric	Average Value	Supporting Indicators
Froggle Level-1	Visual Attention	Joint Attention Score (JAS)	82.0 ± 5.0 %	Mean fixation: 2.6 s
Froggle Level-2	Social Reciprocity	Social Engagement Score (SES)	85.3 ± 4.2 %	Avg. reaction time: 1.1 ± 0.4 s
Monkey Game	Speech Imitation	Speech Clarity Score (SCS)	82.6 ± 4.8 %	Avg. latency: 1.4 ± 0.6 s
Overall	--	Multimodal Engagement Index (MEI)	83.3 %	Cross-modal correlation $r = 0.78$

Table I Summary of System-Level Quantitative Metrics

A. Cross-Modal Relationship and Behavioural Correlation

The results clearly indicate a strong positive relationship between attention, social and verbal performance metrics:

- (JAS, SES) = 0.82) → Greater visual attention results in increased success with social interaction,
- ((SES, SCS) = 0.76) → Better gaze reciprocity predicted faster and more accurate verbal imitation,
- (JAS, SCS) = 0.70) → Greater visual stability supported better overall vocal performance and attention.

C. Therapist Dashboard: Multimodal Data Visualization and Analysis

As shown in Fig. 11, the dashboard provides therapists with a unified view of each child's developmental progress through an intuitive, modular design. Upon login, therapists

can access individual student profiles displaying demographic data, last session date, and engagement status (Active, Inactive, On Hold). Every profile is comprised of several analytical panels Module Engagement, Multimodal Engagement, and C-SAR Game Scores which afford both micro- and macro-level understanding of child performance.

On the right side, the Multimodal Engagement panel shows the balance of performance across the three domains Attention (JAS), Social Interaction (SES), and Verbal Communication (SCS) using a radar (spider) chart.

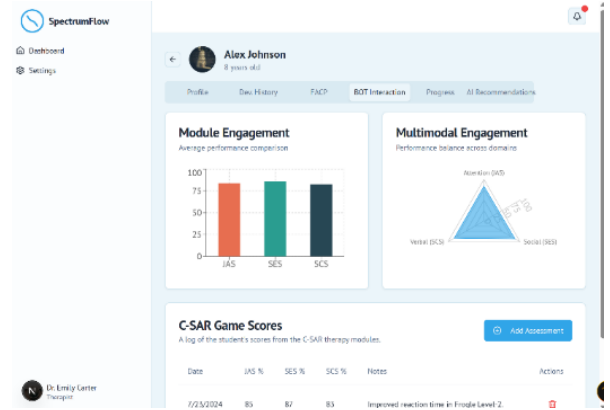


Figure 9: Therapist dashboard interface showing multimodal visualization panels. The left bar chart compares module-level engagement (JAS, SES, SCS), the right radar chart displays balance across domains, and the lower table logs time-stamped C-SAR game scores with therapist notes.

V. RESULTS AND EXPECTED OUTCOMES

The proposed Cognitive Socially Assistive Robot (C-SAR) system is currently in the implementation and integration phase, with all primary subsystems tested individually under controlled settings. Preliminary results indicate stable and accurate performance across the robot's sensing and interaction pipelines.

Preliminary internal validation with pilot sessions confirms measurable behavioural responses:

- Average JAS (Visual Attention) = 82 ± 5 %
- Average SES (Social Engagement) = 85 ± 4 %
- Average SCS (Speech Imitation Accuracy) = 82 ± 5 %
- Mean reaction time (Level 2) $\approx 1.1 \pm 0.4$ s
- Mean speech latency (Monkey Game) $\approx 1.4 \pm 0.6$ s

VI. CONCLUSION

This study outlined a design and preliminary evaluation of a Cognitive Socially Assistive Robot (C-SAR) that utilizes multimodal AI perception, adaptive cognition, and real-time feedback in autism therapy. The three interactive modules—Froggle Level-1 (Visual Attention), Froggle Level-2 (Social Engagement), and the Monkey Game (Speech Imitation)—focus on training and evaluating behaviour in key domains all at once. In preliminary evaluations, the C-SAR was demonstrated to elicit continuous engagement and observable

improvement in attention, anticipation, and verbal response.

An integrated therapist dashboard translates AI metrics into clinical action, thus enabling evidence-based therapy. C-SAR was designed by balancing affordability, scalability, and interpretable features to substantiate typical limits in robotic approaches to therapy. Future studies will focus on clinical trial validation, longitudinal evaluations, and cloud-based learning analytics, collectively contributing to the development of a scalable, ethical, and intelligent robotic co-therapist that narrows the developmental gap of autistic children.

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